

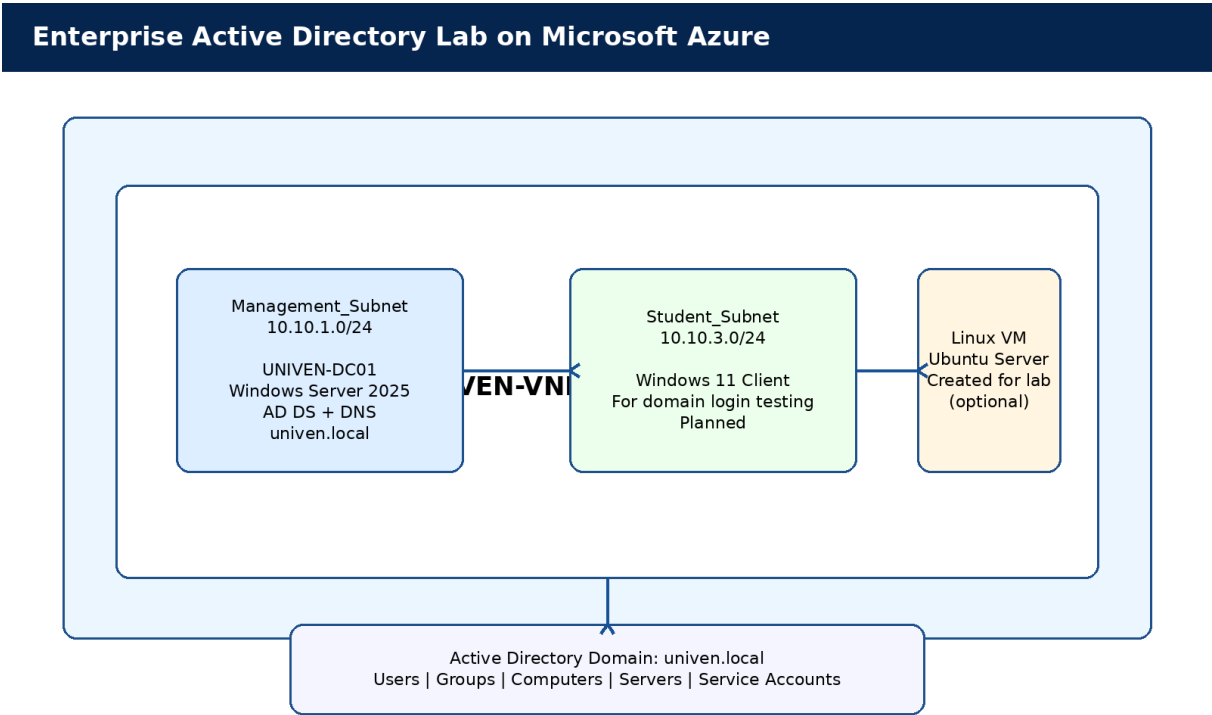
Enterprise Active Directory Domain Services Lab on Microsoft Azure

Author: Mulisa Netshidzivhe
Domain: univen.local
Server: UNIVEN-DC01
Platform: Microsoft Azure + Windows Server 2025

Project Overview

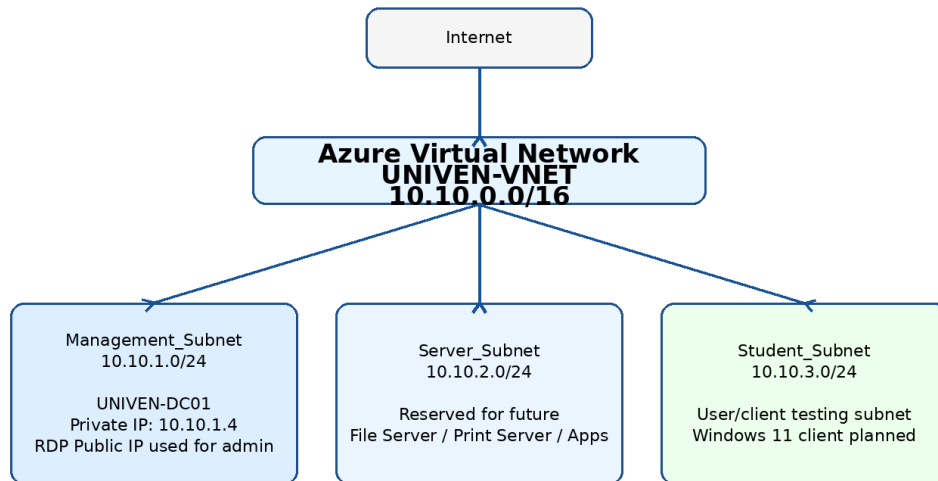
This case study documents a practical enterprise identity lab deployed on Microsoft Azure. The lab simulates a small organization with centralized authentication, DNS, domain users, organizational units, and security groups using Active Directory Domain Services.

Architecture



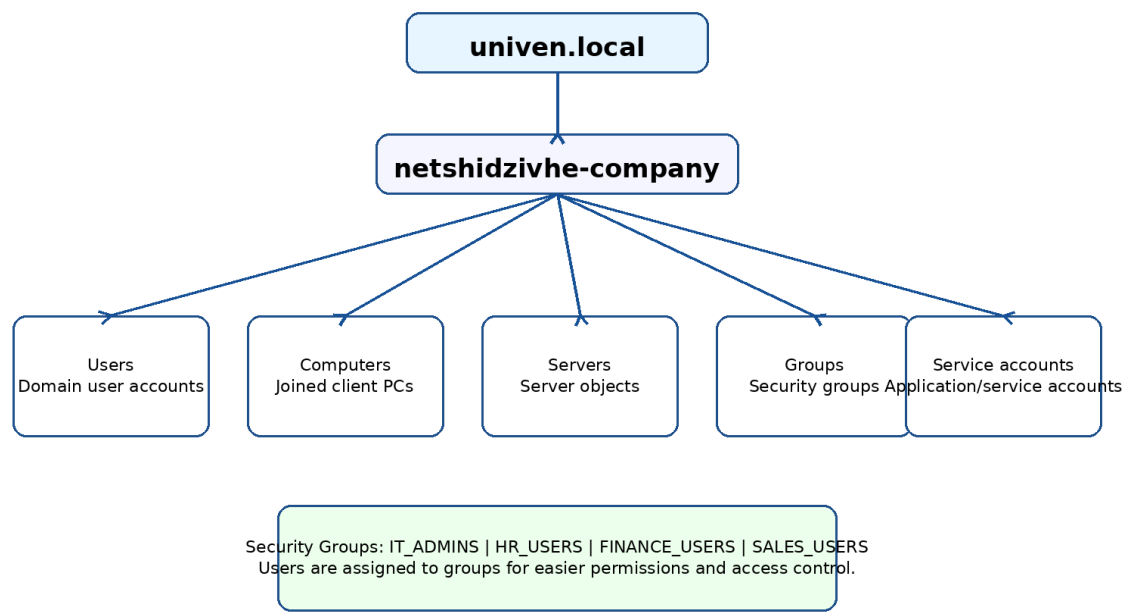
Network Topology

Network Topology



Active Directory OU Design

Active Directory Organizational Unit Design

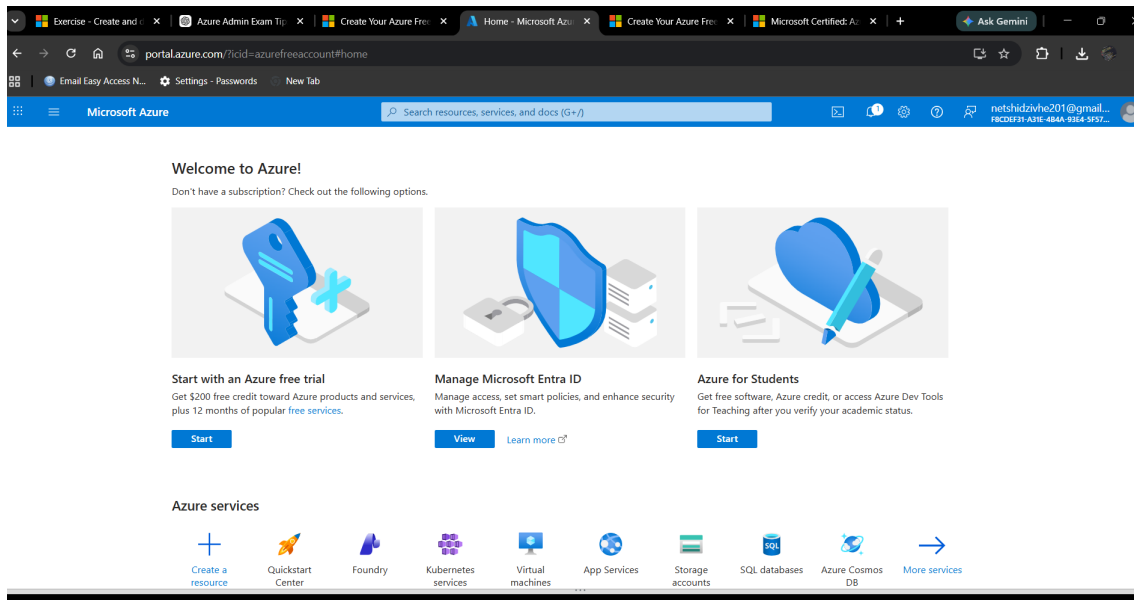


Azure Resources

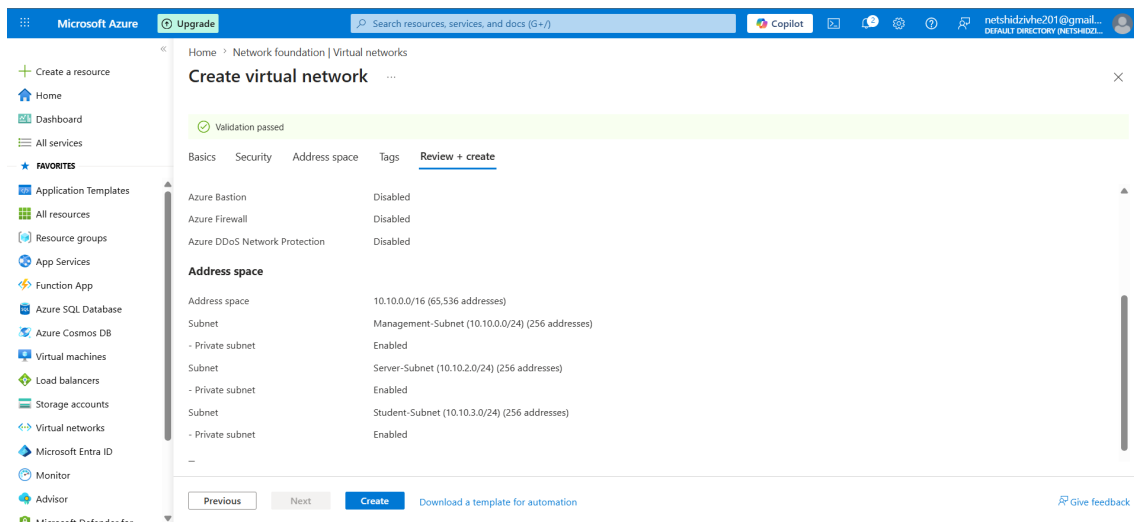
Resource	Purpose
UNIVEN-ICT-RG	Resource group for the lab
UNIVEN-VNET	Virtual network for servers and clients
Management_Subnet	Subnet for Domain Controller
Server_Subnet	Future server workloads
Student_Subnet	Client/user testing subnet
UNIVEN-DC01	Windows Server 2025 Domain Controller

Implementation Evidence - Screenshots

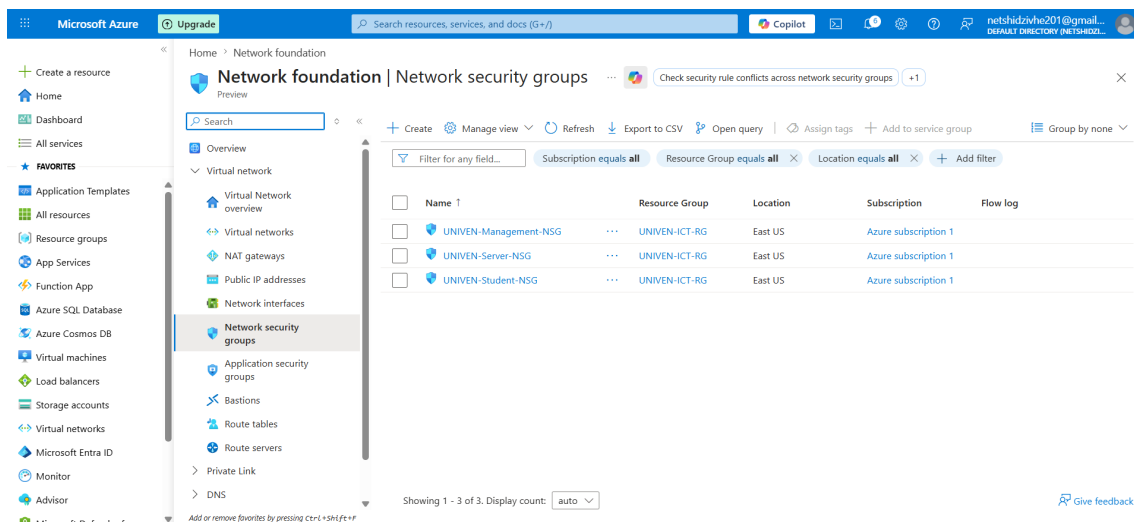
01 Azure Portal Home



03 Vnet Review Create



05 Network Security Groups Created



06 Windows Server Vm Deployment Complete

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

netshidzvh201@gmail...
DEFAULT DIRECTORY NETWORKS...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for

Home

CreateVm-MicrosoftWindowsServer.WindowsServer-202-20260630115516 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-MicrosoftWindowsServ... Start time : 2026/06/30, 12:18:13

Subscription : Azure subscription 1 Correlation ID : 3b8a9478-d741-4c1a-8b6a-955c7...

Resource group : UNIVEN-ICT-RG

> Deployment details

< Next steps

Set up auto-shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource Create another VM Scale out your VM

Cost management

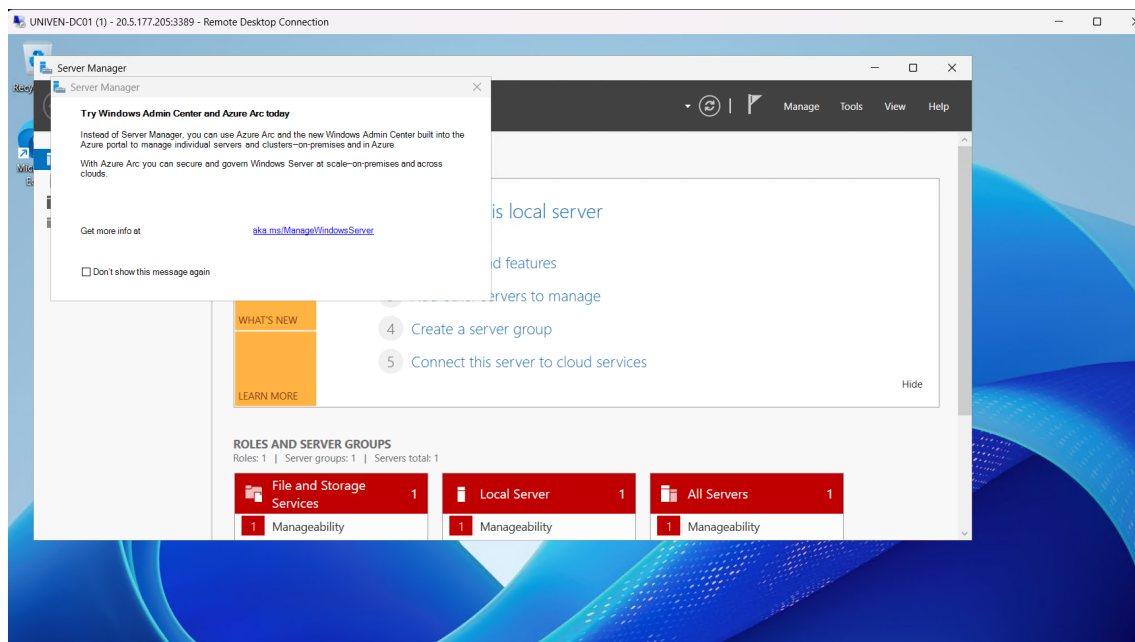
Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

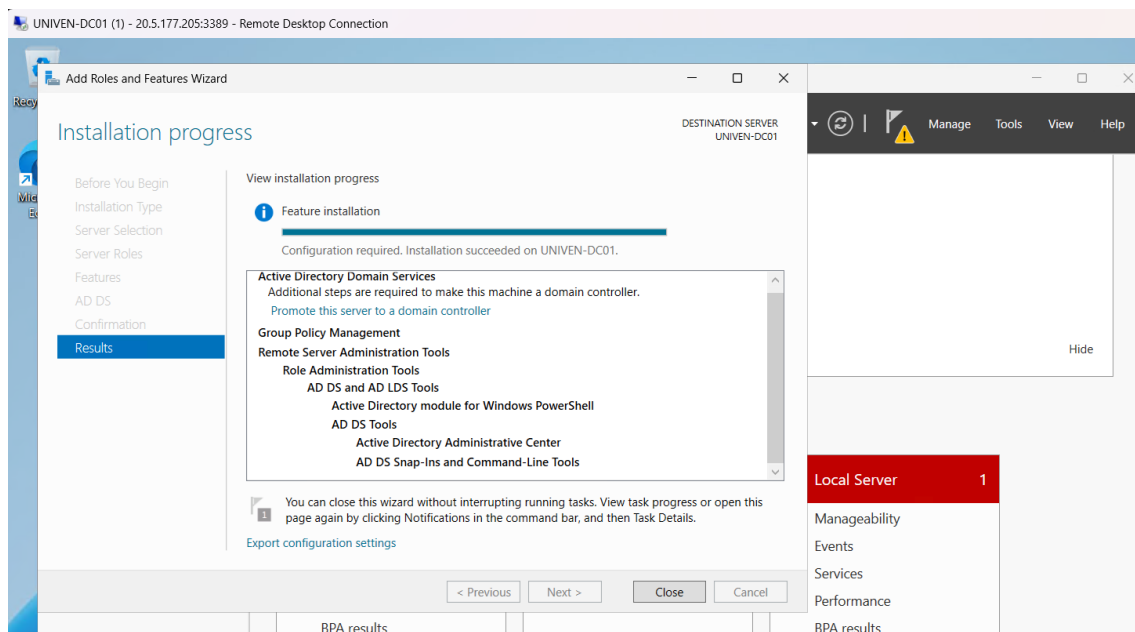
Add or remove favorites by pressing Ctrl+Shift+F

Active Directory Installation and Domain Controller Promotion

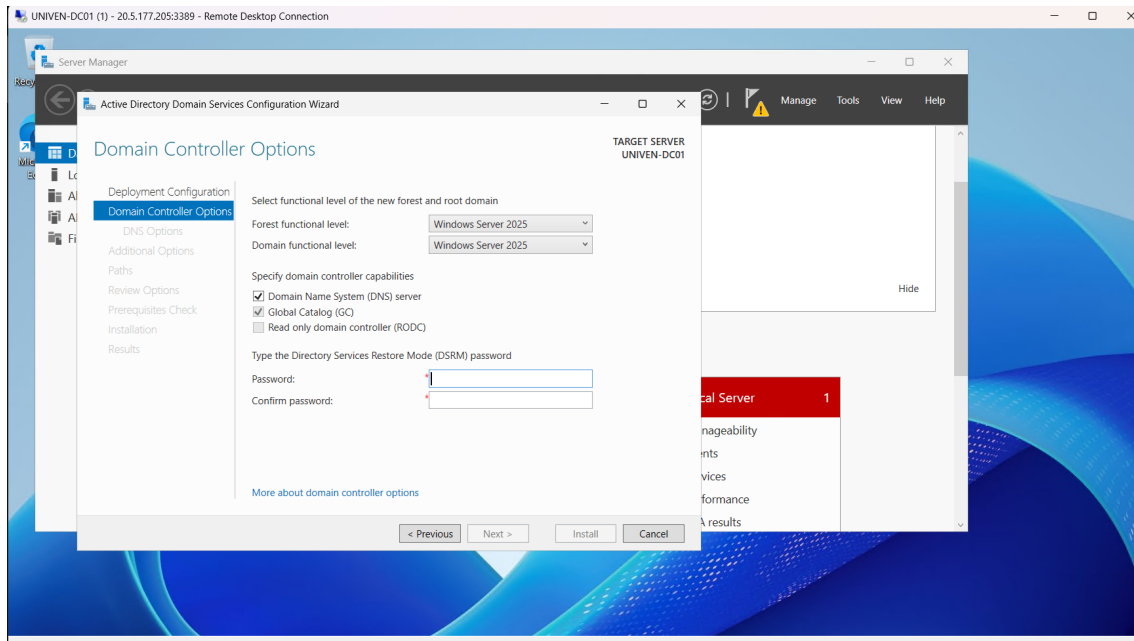
09 Server Manager Dashboard



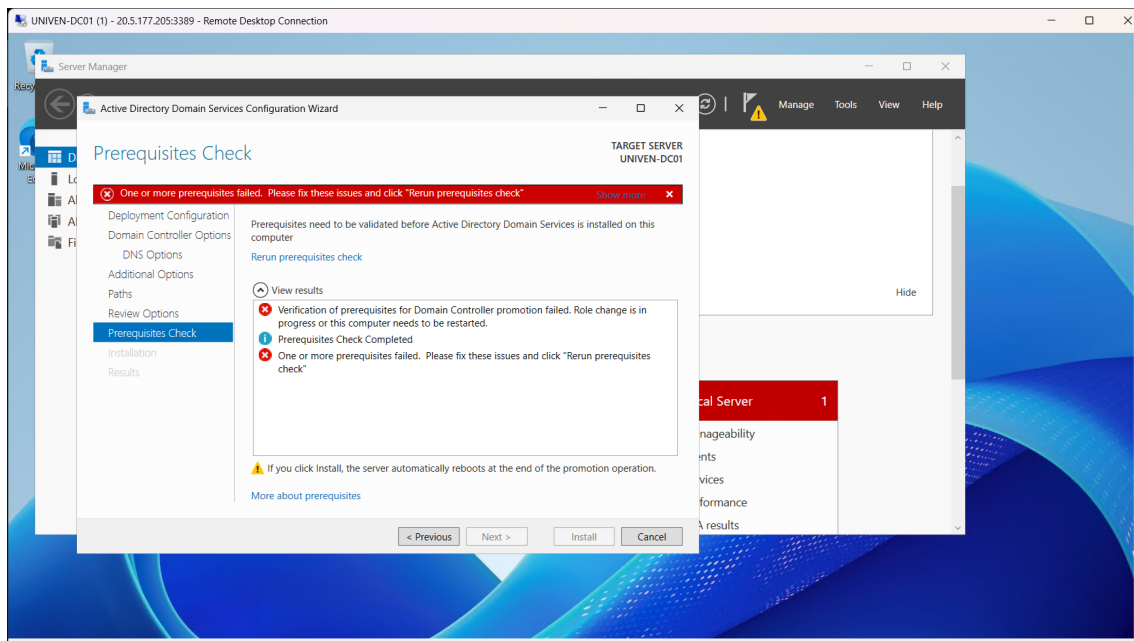
10 Ad Ds Installation Progress



11 Ad Ds Installation Complete



12 Dc Promotion Prerequisite Issue



15 Domain Controller Local Server

UNIVEN-DC01 (1) - 205.177.205:3389 - Remote Desktop Connection

Recycle Bin

Microsoft Edge

Server Manager

Server Manager ▸ Local Server

Manage Tools View Help

Dashboard

Local Server

All Servers

AD DS

DNS

File and Storage Services ▸

PROPERTIES

For UNIVEN-DC01

TASKS

Computer name	UNIVEN-DC01	Last installed updates	Today at 10:24 AM
Domain	univen.local	Windows Update	Never check for updates
		Last checked for updates	Today at 10:23 AM
Microsoft Defender Firewall	Domain: On	Microsoft Defender Antivirus	Real-Time Protection: On
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Enabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC) Coordinated Universal Time
Ethernet	IPv4 address assigned by DHCP, IPv6 enabled	Product ID	00493-70000-00001-AA5
Azure Arc Management	Disabled		
Remote SSH Access	Enabled		

Operating system version	Microsoft Windows Server 2025 Datacenter Azure Edition	Processors	AMD EPYC 7763 64-Core Processor
Hardware information	Microsoft Corporation Virtual Machine	Installed memory (RAM)	3.00 GB

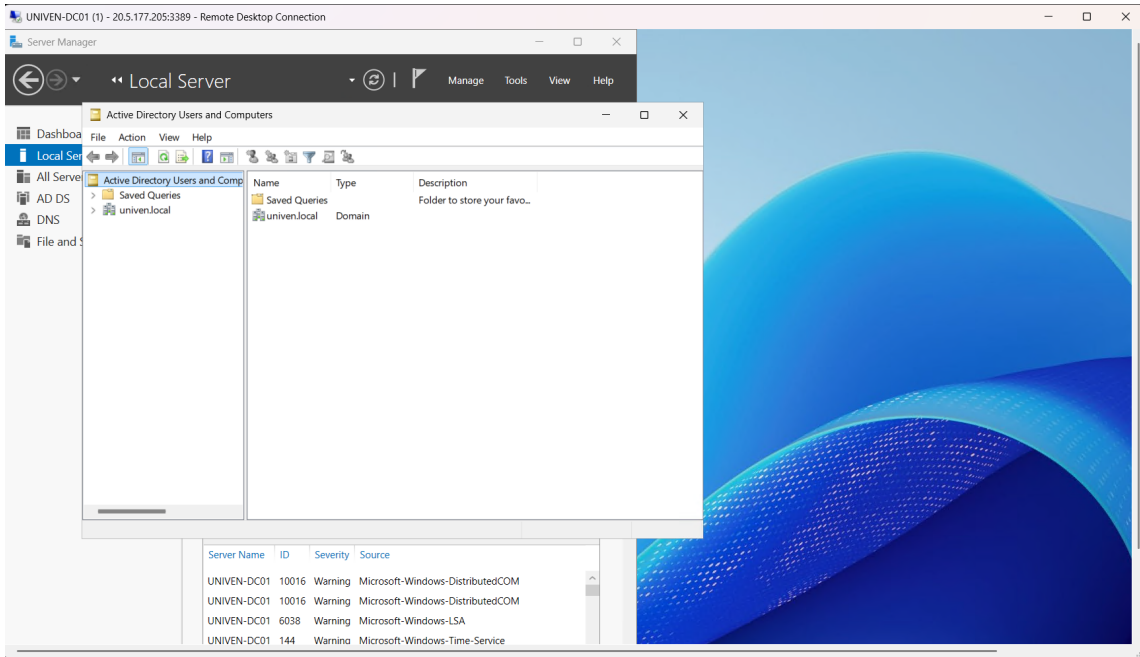
EVENTS

All events | 50 total

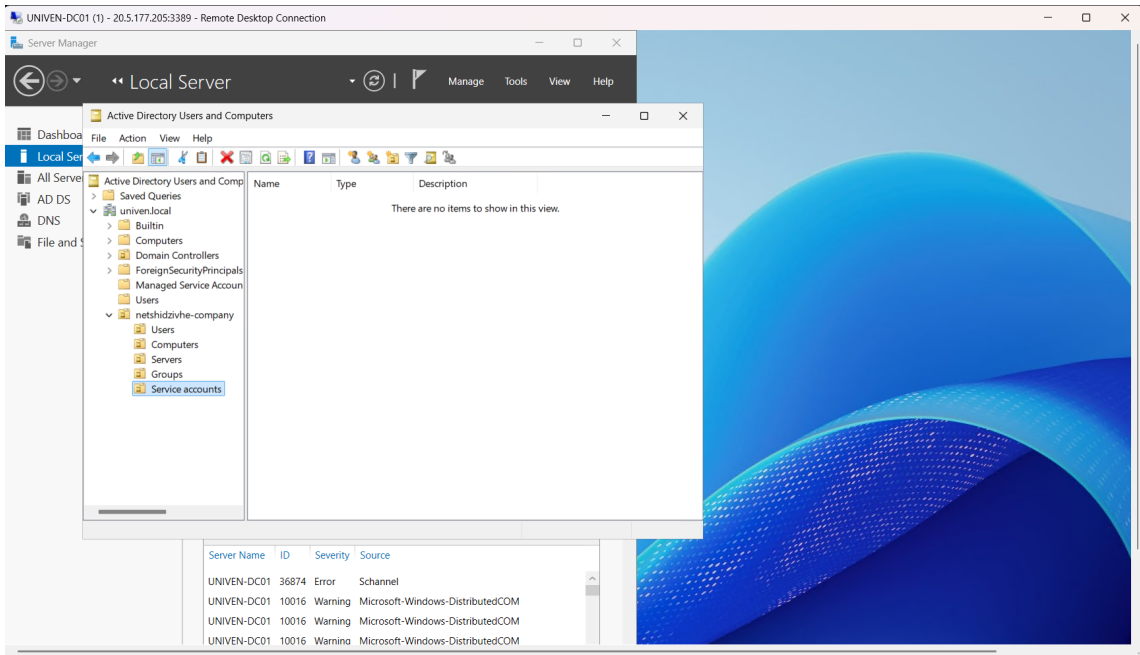
TASKS

Active Directory Administration Evidence

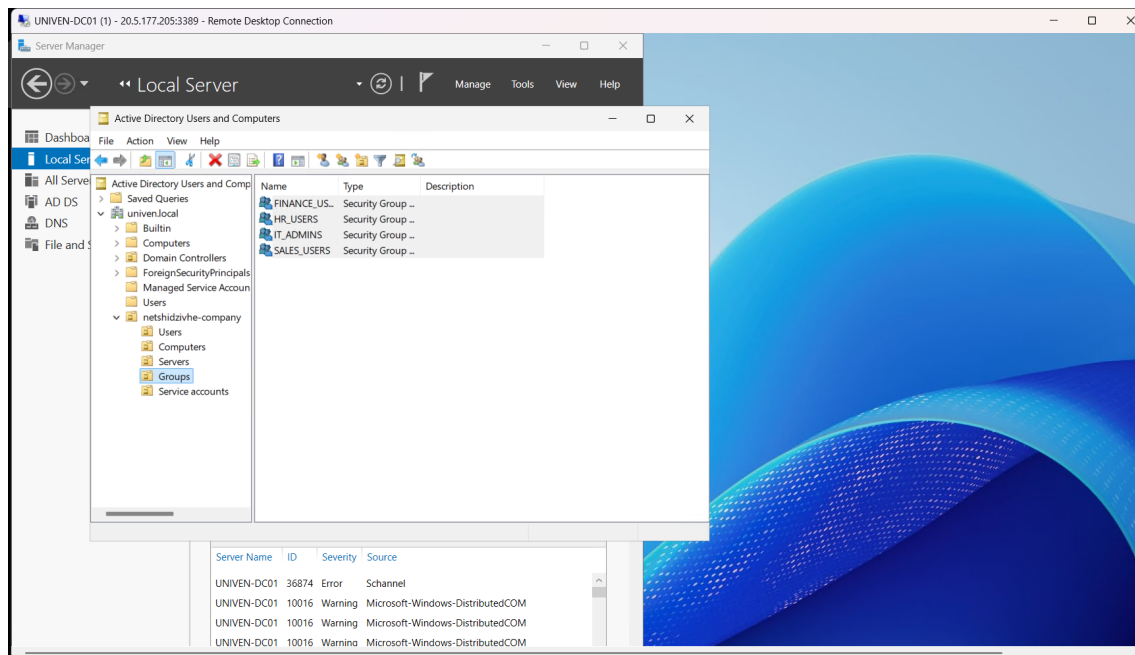
16 Active Directory Domain Visible



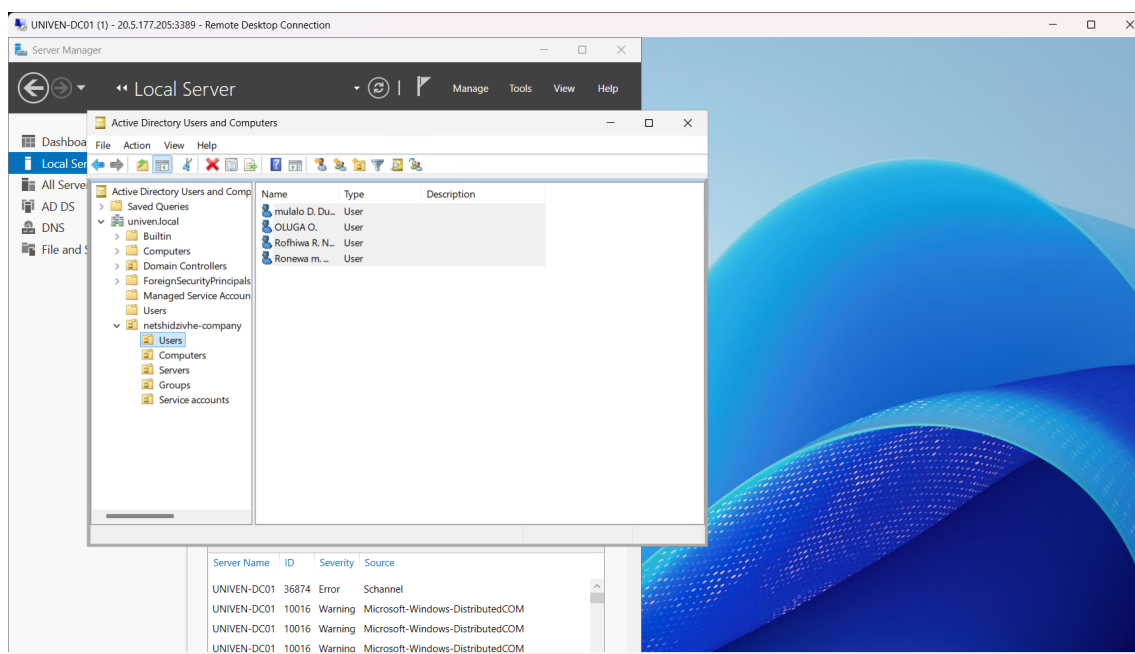
17 Active Directory Ou Structure



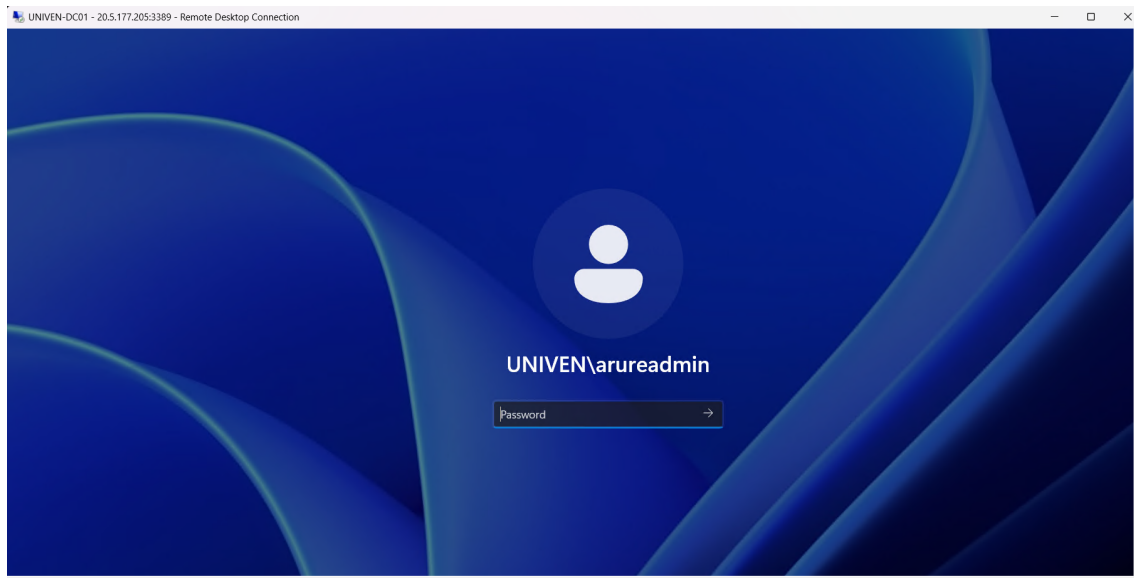
18 Security Groups Created



19 Users Created



21 Domain Sign In Screen



Challenges and Solutions

Challenge	Solution
AD DS prerequisite check failed	Restarted the server and reran the promotion wizard.
VM size availability problems	Changed region and selected an available VM size.
Windows 11 client VM blocked	Continued server-side testing and prepared for future client testing.

Skills Demonstrated

Azure administration, Windows Server administration, Active Directory Domain Services, DNS, Organizational Unit design, security group management, identity and access management, RDP troubleshooting, infrastructure documentation, and PowerShell automation.

Conclusion

The project demonstrates hands-on deployment of an enterprise identity infrastructure using Microsoft Azure and Windows Server. It provides practical evidence of skills relevant to IT Support, Systems Administration, Azure Administration, and Identity and Access Management roles.